Identifying the Causes of Protectionism using Double Machine Learning

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1 Proposed Topic

Why do some democracies protect domestic producers more than others? I aim to study whether factors such as exports, unemployment rates, GDP, inflation rates, and the number of political parties causally shape how protectionist a country's trade policy is. Using a panel of many countries spanning several decades, I will estimate the effect of each factor using double machine learning (DML), which lets flexible learners such as random forests and neural networks estimate causal parameters in the presence of complex confounding variables (Chernozhukov et al., 2016).

Research Question: *Which domestic factors cause countries to adopt more protectionist trade policy, and how robust are these effects?*

This analysis aims to build upon prior analyses such as (Ehrlich, 2007) and similar, CITE. Prior analyses have used linear regression to evaluate the causes of protectionism. I propose to use double machine learning to validate the results of (Ehrlich, 2007) and similar, CITE, creating a second avenue to test causal relationships in political science.

This analysis aims to test the following hypotheses: write a little more on why each is expected and why alternate would not be expected

> **H1 (Exports).** Exports being a higher percentage of GDP increases/decreases – which makes more sense? protectionism.

> **H2 (Unemployment).** Increased unemployment rates cause increased protectionism.

> **H3 (GDP).** Increased GDP per capita and total GPD in US$ I have data for both – combine or separate into two hypotheses? cause increased or decreased? protectionism.

> **H4 (Inflation).** Increased inflation rates cause increased protectionism.

> **H5 (Number of Political Parties).** More political parties causes increased protectionism

For the sake of this analysis, the level of protectionism will be quantified using tariff rates explain why this is a reasonable thing to do. In other words, higher tariff rates mean higher levels of protectionism, and vice versa.

2 Proposed Sources

2.1 Access to Protection: Domestic Institutions and Trade Policy in Democracies (Ehrlich, 2007)

Sean Ehrlich's 2007 paper "Access to Protection: Domestic Institutions and Trade Policy in Democracies" tests various hypotheses regarding the causes of protectionism. He hypothesizes that more electoral districts, more political parties, low party discipline, bicameralism, presidentialism, and left-wing governments all lead towards increased protectionism. Using linear regression, Ehrlich determined that all hypotheses were true, except for the hypothesis that bicameralism causes increased protectionism, which was found false. What about low party discipline? Not in class notes –> check paper itself. Ehrlich also discusses access points, which refer to a policymaker susceptible to lobbying. Ehrlich argues that an increase in access points leads to an increase in protectionism.

2.2 Possibly: Trade Protectionism: Reasons and Outcomes (Abboushi, 2010)

2.3 Another similar paper to Ehrlich, find one

2.4 Double ML paper (Chernozhukov et al., 2016), also maybe python documentation

2.5 Possibly: Why Trade, and What Would be the Consequences of Protectionism? (Jean and Reshef, 2017)

3 Existing Analysis

More in depth summary of Ehrlich2007 and similar papers

4 Proposed Data and Methods

Data on tariff rates (World Bank Group, 2026*e*), exports (World Bank Group, 2026*a*), unemployment rates (World Bank Group, 2026*f*), GDP (World Bank Group, 2026*b*,*c*), and inflation (World Bank Group, 2026*d*) were obtained from World Bank Group, and data on the number of political parties (Varieties of Democracy, 2026) was obtained from Varieties of Democracy Re-word this, sounds ridiculous since all the citations show where the data comes from. The required data reduction and cleaning is complete. A simple Python program CITE and/or link to my GitHub was written to pull the data needed from each dataset, and add the data to one large .csv file. A subset of the data is shown in table 1.

Country	Year	% Exports	Unemp.	GDP/Capita	GDP (US$)	Inflation	No. Parties	Tariff
ALB	1997	10.04	16.874	728.55	4.71e9	33.18	2.87	14.41
ALB	2000	19.53	19.023	1160.42	6.15e9	0.05	2.87	11.65
...	...	...	...	...	...	...	...	
ZMB	2020	46.72	6.032	951.64	2.34e10	15.73	2.78	4.82

Table 1. *Dataset that will be used for double machine learning. Country and Year are for reference though it may be interesting to look at each country's level of protectionism over time, year, % exports, unemployment rate, GDP/capita, GDP in US$, inflation rate, and number of political parties are independent variables, tariff rate is the dependent variable.*

Explain DML in detail, say how it works, which algorithms specifically, why those algorithms (likely random forest and neural networks)

5 Preliminary Results

6 Impact of Protectionism

7 Conclusion

8 Notes

- Check DML Paper citation –> May be linking to ArXiv, find actual publication if so
- Is JeanReshef2017 peer reviewed? Double check.
- Check years for World Bank data and V-Dem data
- This will be more than 3 pages –> check with Prof. Toloudis.
  - Maybe can cut sections on impact of protectionism and causes of protectionism since those will be summarized throughout section 2.

References

Abboushi, Suhail. 2010. "Trade protectionism: reasons and outcomes." *Competitiveness Review: An International Business Journal* 20: 384–394.

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, and Whitney Newey. 2016. "Double Machine Learning for Treatment and Causal Parameters." : Forthcoming.

Ehrlich, Sean D. 2007. "Access to Protection: Domestic Institutions and Trade Policy in Democracies." *International Organization* 61(3): 571–605.

Jean, Sébastien, and Ariell Reshef. 2017. Why Trade, and What Would Be the Consequences of Protectionism? Technical report.

Varieties of Democracy. 2026. "Country-Party-Date: V-Party." Varieties of Democracy.

World Bank Group. 2026*a*. "Export of goods and services (% of GDP) [NE.EXP.GNFS.ZS]." World Development Indicators.

World Bank Group. 2026*b*. "GDP (constant 2015 US$) [NY.GDP.MKTP.KD]." World Development Indicators.

World Bank Group. 2026*c*. "GDP per capita (current US$) [NY.GDP.PCAP.CD]." World Development Indicators.

World Bank Group. 2026*d*. "Inflation, consumer prices (annual %) [FP.CPI.TOTL.ZG]." World Development Indicators.

World Bank Group. 2026*e*. "Tariff rate, applied, weighted mean, all products (%) [TM.TAX.MRCH.WM.AR.ZS]." World Development Indicators.

World Bank Group. 2026*f*. "Unemployment (% of total labor force) [SL.UEM.TOTL.ZS]." World Development Indicators.